

HealthGuard AI

Healthcare Software Requirements Specification

Clinical Data Management & Compliance Automation System

Document Information	Details
Product Name	HealthGuard AI
Document Version	2.0
Release Date	October 21, 2025
Document Status	Approved for Development
Classification	FDA Class II Medical Device Software
IEC 62304 Class	Class B (Moderate Risk)

Executive Summary

HealthGuard AI is a comprehensive compliance testing automation platform designed specifically for healthcare software development. This Software Requirements Specification (SRS) document defines the functional, non-functional, data, and interface requirements for HealthGuard AI version 2.0.

The system is designed to meet stringent regulatory requirements including:

- FDA 21 CFR Part 11 (Electronic Records; Electronic Signatures)
- HIPAA Privacy and Security Rules (45 CFR Parts 160, 162, and 164)
- IEC 62304:2006/AMD1:2015 (Medical Device Software Life Cycle Processes)
- ISO 13485:2016 (Medical Devices Quality Management Systems)
- ISO 14971:2019 (Application of Risk Management to Medical Devices)

Functional Requirements - Authentication

REQ-AUTH-001: Multi-Factor Authentication (MFA)

Priority: Critical | **Category:** Functional - Security

The system shall require multi-factor authentication for all user accounts accessing patient health information (PHI) or system administration functions. MFA must combine at least two of the following factors: knowledge factor (password), possession factor (mobile device, security token), or inherence factor (biometric).

Compliance Standards:

- HIPAA §164.312(a)(2)(i) - Access Control - Authentication
- FDA 21 CFR Part 11.300(b) - Password Requirements
- NIST SP 800-63B - Digital Identity Guidelines

Document Overview

This PDF contains a summary of the comprehensive healthcare requirements document. The complete document includes 25+ detailed functional requirements covering:

- User Management & Authentication (5 requirements)
- Patient Data Management (4 requirements)
- Clinical Workflows (2 requirements)
- Audit & Compliance (1 requirement)
- Interface Requirements (1 requirement)
- Non-Functional Requirements (3 requirements)
- Data Requirements & Privacy Controls
- Compliance Mapping Matrix
- Risk Analysis & Mitigation (20+ hazards)
- Traceability Matrix
- Validation Requirements

Total Test Cases: 150+ detailed test scenarios

Compliance Standards: FDA, HIPAA, IEC 62304, ISO 13485, ISO 14971

Document Pages: ~40 pages (when fully formatted)

Word Count: ~23,000 words